

3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

A High-Performance Handheld Gas Chromatograph

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ABSTRACT

Gas chromatography is a prominent technique for separating complex gasses and analyzing relative quantities of each separate component in the blend. This analytical technique is popular with scientists in a wide range of applications, including air and water pollution and chemical and biological analysis. Today, one of the hot topics in the analytical instrumentation community is the need to move analysis away from the laboratory to where samples ("the field") are. The need to do this is fueled by the need for quicker real time data collection and lower analysis cost. Our recent improvement on a Handheld Gas Chromatograph is developing a sample pre-concentrator unit and related electronics to improve the detection capability of trace chemical components in air. The hand-held, real time detection gas chromatograph (GC) prototype unit was developed earlier at the Micro-Technology Center of Lawrence Livermore National Laboratory by means of Micro-Electro-Mechanical-System (MEMS) technology. The total weight of this hand-held gas chromatograph is about 8 lbs. and its physical size is 8" x 5" x 3". It consumes about 12 watts of electrical power and has a response time on the order of two minutes. At present, the new detector is a Glow Discharge detector with a sensitivity of parts per billion. The average retention time is about 30 to 45 seconds. Under optimum condition, the calculated effective plate number for the MEMS silicon column is about 40K. The separation column in the portable GC is fabricated completely on silicon wafers. Silicon is a good thermal conductor and it provides a uniform temperature for the whole column together with the possibility of rapid heating and cooling of the column. The operational temperature can be as high as 350 degrees Celsius (maximum column coating temperature). The GC system is capable of rapid column temperature ramping and cooling operations. These capabilities are especially important for organic and biological analyses in GC applications.